03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 11

(JCD 8.1 - 8.3)

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Previous lecture

- Convective regions are mass-dependent:
  - Massive stars: convective core, radiative envelope.
  - Intermediate stars: convective envelope and core with radiative zone in between.
  - Low-mass stars: Convective envelope and radiative core.
  - Coolest, lowest-mass stars: Fully convective
- M-L relation also mass-dependent:
  - Lower Main sequence: $L_{*,\text{low mass MS}} \propto M_*^5 \mu^{7.5}$
  - Upper main sequence: $L_{*,\text{high mass MS}} \propto M_*^3 \mu^4$
  - Fully convective stars: $L_{*,\text{convective}} \propto M_*^{-7} T_{\text{eff}}^{40}$

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Outline

Nuclear energy generation

Quantum tunneling

Cross sections

Energy release - binding energy

Summary

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Nuclear energy generation

- During most of a star's life, energy is generated via nuclear reactions [remember lecture 1]
- Due to these reactions, the chemical composition in stars changes with time, eg for the hydrogen mass fraction:

$$\frac{dX}{dt} = r_X$$

- Remember lecture 8: We want to find an expression for the energy generation rate per unit mass ϵ. This depends on:
  - Cross section for a reaction
  - Energy released per reaction
  - Total reaction rate

Nuclear reaction notation

- There are several ways in which reactions can be written.
- Writing it out fully, the particles / atoms that react are on the left and the resulting particles / atoms are on the right, eg:

$$A \quad + \quad a \quad \rightarrow \quad Y \quad + \quad y$$

- Common used shorthand is

$$A(a, y)Y$$

- Note that this could easily also be $A(a, Y)y$, $a(A, Y)y$, or $a(A, y)Y$, but convention is to place the primary particles outside the brackets

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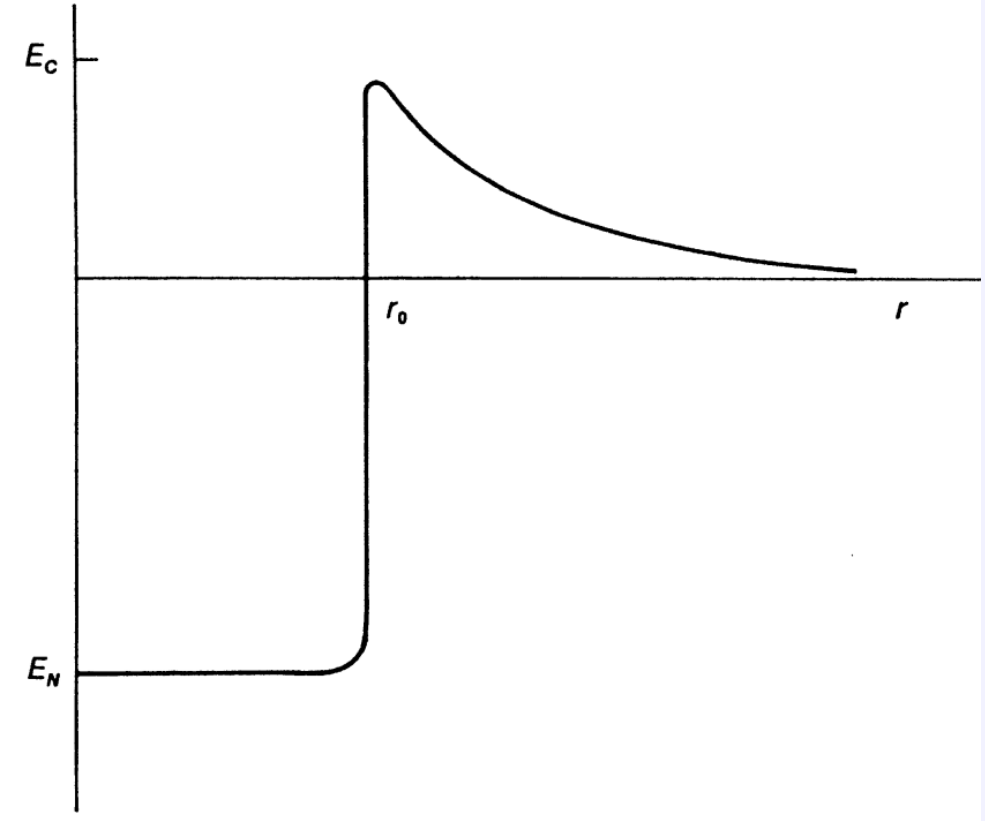
Overcoming Coulomb barrier

- Nuclei are held together by the strong force
- For nuclei to react, they need to overcome the Coulomb repulsion
- The approaching nucleus has thermal kinetic energy



(Figure credit: JCD)

Overcoming Coulomb barrier - temperature required

- Assume 2 particles with atomic numbers Z_1, Z_2 and charges q_1, q_2
- The Coulomb energy is then

$$E_C = \frac{q_1\, q_2}{4\,\pi\,\epsilon_0\, r} = \frac{Z_1\, Z_2\, e^2}{4\,\pi\,\epsilon_0\, r}$$

- Thermal kinetic energy is typically

$$E_{kin} = E_{th} = \frac{3}{2}\, k\, T$$

- Equating both energies, gives you the temperatures required for the nuclear reactions to start:

$$T = \frac{2\, Z_1\, Z_2\, e^2}{12\,\pi\,\epsilon_0\, k\, r}$$

Coalescing two protons

- Proton radius: $r_p = 8.4 \cdot 10^{-16}\,\mathrm{m}$
- For two protons, $Z_1 = Z_2 = 1$
- Proton charge: $e = 1.6 \cdot 10^{-19}\,\mathrm{C}$
- Vacuum permittivity / electric constant: $\epsilon_0 = 8.85 \cdot 10^{-12}\,\mathrm{C}^2\,\mathrm{J}^{-1}\,\mathrm{m}^{-1}$
- Boltzmann constant: $k = 1.4 \cdot 10^{-23}\,\mathrm{J\,K}^{-1}$
- Then it follows that the temperature required is

$$T \approx 1.3 \cdot 10^{10}\,\mathrm{K}$$

- Stellar interiors have a temperature of about $2 \cdot 10^7\,\mathrm{K}$
- Now what?

Quantum tunneling

- It is clear that classical mechanics does not work. Quantummechanical effects are required.
- Particle thermal energy is a distribution (coming from the Maxwell-Boltzmann velocity distribution)
- Even with a tail of high-energy particles, it is not enough.
- But: small but non-zero probability of penetrating the Coulomb barrier, i.e. quantum tunneling effect
- In a stellar core, at $T \sim 2 \cdot 10^7$ K, about 10^{-8} of all the protons have sufficient energy (fusion is slow process)
- Once they do, the strong nuclear force can fuse the protons together

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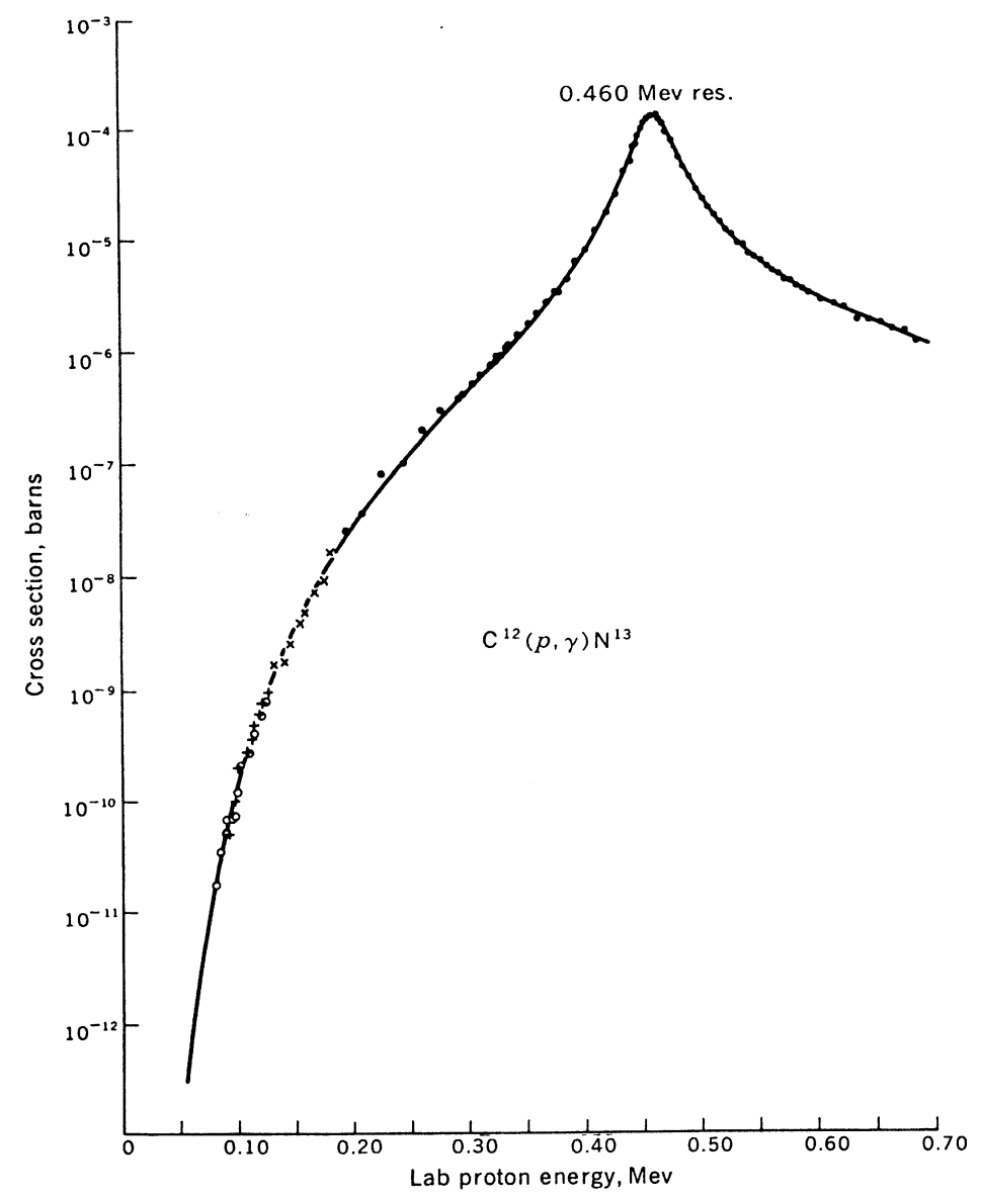
Cross sections

- We need an expression for the cross section of a particular reaction.
- Remember lecture 8 where we introduced cross section
- Can we measure these cross sections, as a function of energy, $\sigma(E)$, in the lab?

Measured cross sections - stellar interior energy $\sim 20\,\mathrm{keV}$



(Figure credit: JCD)

Re-express cross section

- $\sigma(E)$ is related to the probability to tunnel through the Coulomb barrier

$$\exp\left(-\frac{2\pi\, Z_1\, Z_2 e^2}{\hbar\, v\, 4\pi\epsilon_0}\right),$$

  where $\hbar = h/2\pi$ is the reduced Planck's constant and v the relative speed between nuclei

- $\sigma(E)$ is also related to the geometrical cross section of the nuclei:

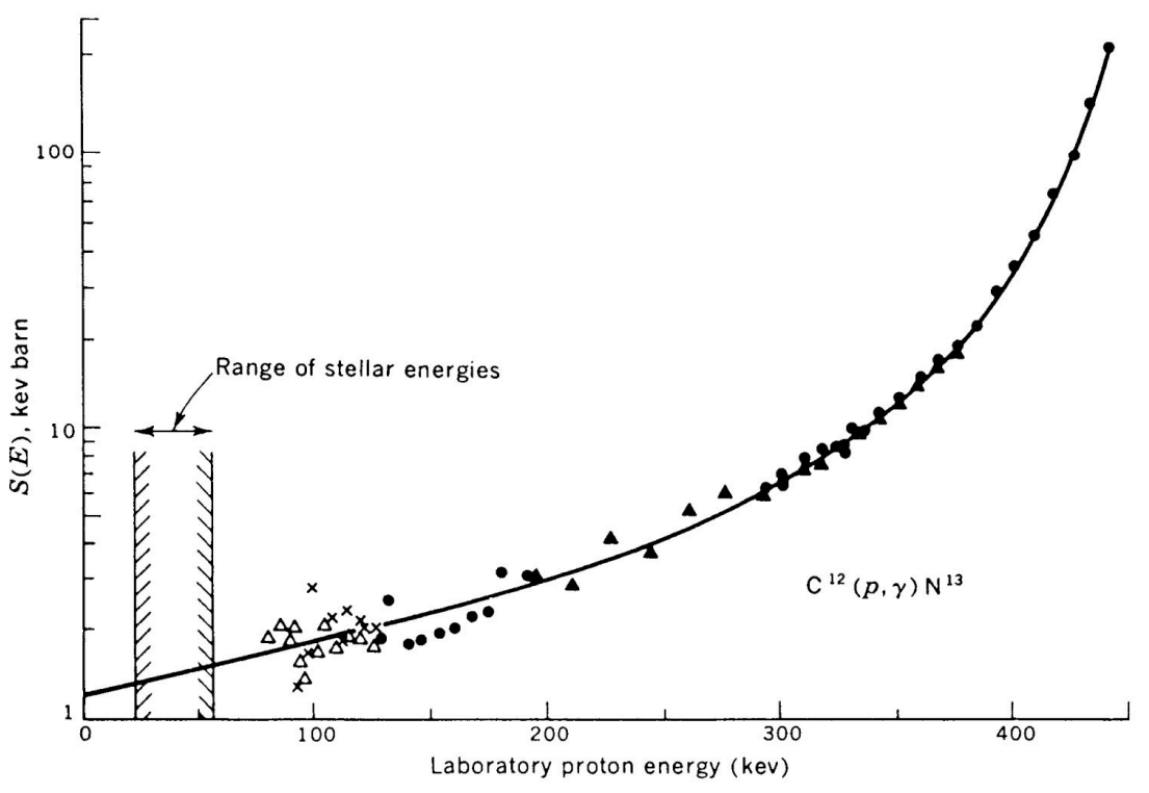
$$\sigma_{geom}(E) = \pi\,\lambda^2 \propto \frac{1}{p^2} \propto \frac{1}{E},$$

  with λ the de Broglie wavelength and p the momentum of the nucleus.

Astrophysical cross section factor

- We can now define a new factor $S(E)$ such that

$$\sigma(E) = \frac{S(E)}{E} \exp\left(-\frac{2\pi\, Z_1\, Z_2 e^2}{\hbar\, v\, 4\pi\epsilon_0}\right)$$



(Figure credit: JCD)

Gamow energy

- The energy E of the particles is measured in the centre-of-mass system and related to their relative speed v:

$$E = \frac{1}{2} m\, v^2 = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} v^2 = \frac{1}{2} \frac{A_1 A_2}{A_1 + A_2}\, m_u\, v^2 = \frac{1}{2} A\, m_u v^2,$$

  where m is the reduced mass and A the reduced atomic weight of the nuclei.

- We can now write the cross section as

$$\sigma(E) = \frac{S(E)}{E} \exp\left(-\frac{b}{\sqrt{E}}\right),$$

  where $E_G = b^2$ is the Gamow energy.

- We can show that the Gamow energy E_G is:

$$\sqrt{E_G} = \frac{2\pi\, e^2 \sqrt{m_u}}{\sqrt{2}\hbar\, 4\pi\, \epsilon_0} Z_1\, Z_2\, \sqrt{A}$$

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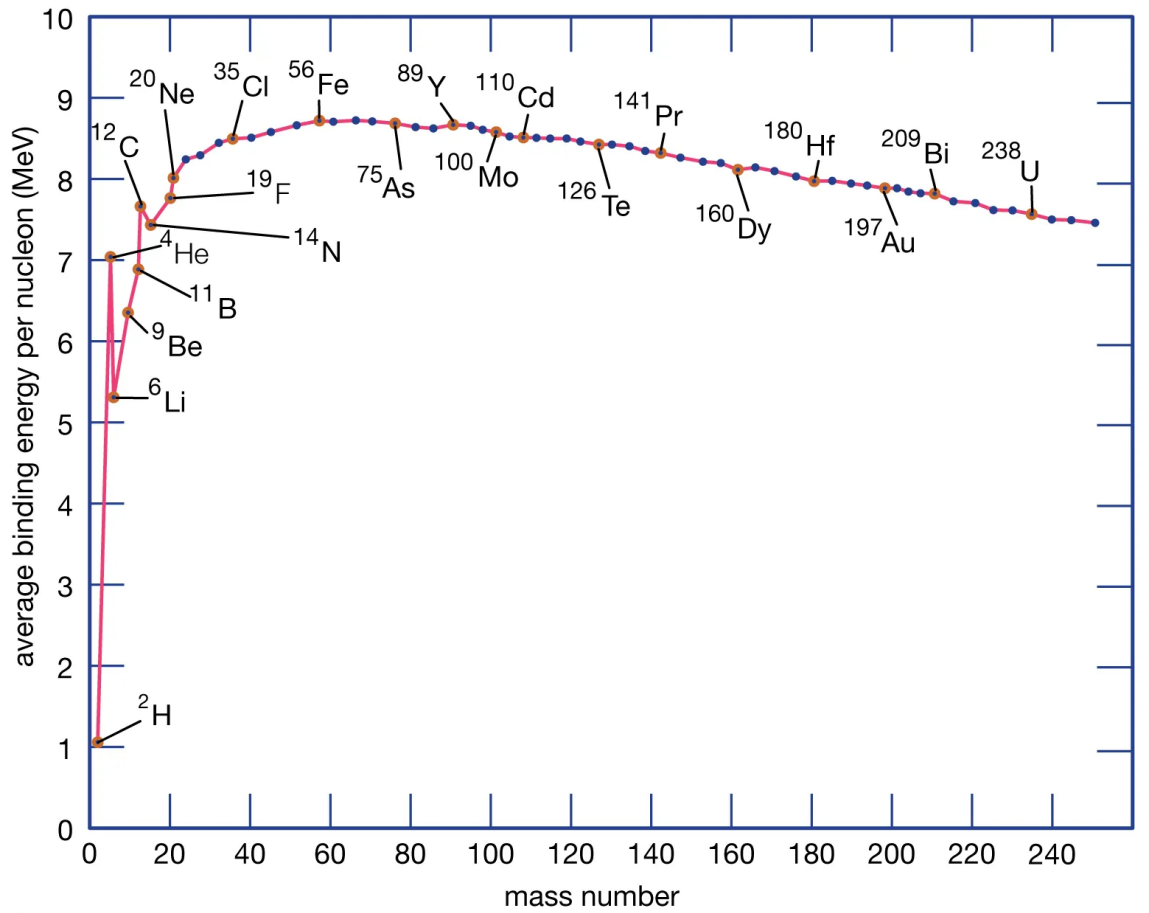
Energy release

- Mass - energy equivalence $E = m\,c^2$ means that we can use the difference in mass between particles before and after the reaction to calculate the release of energy.
- Assume again our reaction $A(a, y)Y$. The energy released will be
$$Q = [m(A) + m(a) - m(Y) - m(y)]\,c^2$$
- It is more convenient to think in mass excess Δm between particle mass and the mass from the sum of its protons (Z) and neutrons (N):
$$\Delta m = m - (Z + N)m_u$$
- The released energy is then
$$Q = [\Delta m(A) + \Delta m(a) - \Delta m(Y) - \Delta m(y)]\,c^2$$

Binding energy

- Binding energy is the energy required to break up the nucleus into nucleons
- The released energy in nuclear reactions is thus closely related to the binding energy



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Summary

- For nuclear fusion to happen, quantum tunneling is required
- We defined the cross section of a reaction in terms of energy:

$$\sigma(E) = \frac{S(E)}{E} \exp\left(-\frac{b}{\sqrt{E}}\right),$$

- The Gamow energy E_G is defined as

$$\sqrt{E_G} = b = \frac{2\pi\, e^2 \sqrt{m_u}}{\sqrt{2}\hbar\, 4\pi\, \epsilon_0} Z_1\, Z_2\, \sqrt{A}$$

- The released energy (related to nuclei binding energy) is

$$Q = [\Delta m(A) + \Delta m(a) - \Delta m(Y) - \Delta m(y)]\, c^2$$

NEXT LECTURE: reaction rate and common nuclear reactions